Viyog: Separating Adversarial and Out-of-Distribution

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***Abstract*—Safety-critical systems must respond differently to two anomalies: *out-of-distribution* (OOD) inputs, which call for abstention, and *adversarial* (ADV) attacks, which demand rejection. Existing detectors collapse both into a single anomaly score. We present *Viyog*, a training-free plug-in second stage that separates OOD from ADV by reading the *dormant-band roughness* of the first convolutional activation—how spatially jagged its quietest channels are (a magnitude-normalised *total variation*, i.e. average change between neighbouring pixels), computable from the forward pass already in progress and storing only ≈ 0.3 KB ($O(C)$) versus 4.5–40 MB for feature-distance baselines.**

Gradient attacks inject broadband high-frequency residue into the quietest first-layer channels; natural inputs (ID and OOD alike) leave them smooth. Viyog exploits this asymmetry: it detects adversarials at AUROC 0.966 (where 1.0 is perfect and 0.5 is chance) while logit detectors are blind (≤ 0.41); combined with logit scores it gives the first deployable three-way ID/OOD/ADV separator (balanced accuracy 0.798 over 20 architectures; 0.863 on GTSRB). We prove scale-invariance (P1)—so norm-preserving attacks achieve 0% evasion—and a concrete separation bound (P2/P3). The detector generalises across 3 datasets, 20 architectures, 4 attacks, and 1D non-vision signals, adds no parameters, and runs in a single forward pass.

***Index Terms*—Out-of-distribution detection, adversarial robustness, feature norms, deep neural networks, safety-critical systems, embedded AI, architecture generalization.**

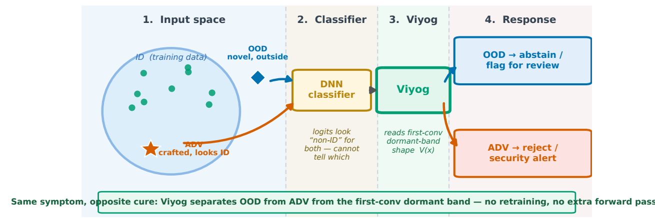

Fig. 1: **The core problem: OOD and ADV inputs require opposite responses, yet both fool the classifier. (1) Input space:** an OOD (out-of-distribution) input lies outside the training cluster; an adversarial (ADV) input is deliberately crafted to sit *inside* the cluster while forcing a wrong prediction. **(2) Classifier:** both produce anomalous logits that existing output-side detectors cannot distinguish. **(3) Viyog:** reads the dormant-band roughness $V(x)$ of the first convolutional activation map—a gradient-free, 0.3 KB statistic—and separates the two. **(4) Response:** OOD $\rightarrow$ abstain / flag for review; ADV $\rightarrow$ reject / security alert. Conflating them wastes adversarial defences on benign inputs or silently accepts attacks.

I. Introduction

Deep neural networks (DNNs) achieve state-of-the-art performance across safety-critical domains including autonomous driving, medical imaging, and embedded vision systems. Deployed systems must therefore detect and respond appropriately to abnormal inputs before acting on model predictions—a requirement that standard classification pipelines do not satisfy [1].

Two kinds of anomalous input require *opposite* responses (Fig. 1): *OOD* inputs fall outside the training distribution and call for abstention, while *adversarial* (ADV) inputs are crafted to force misclassification and demand active rejection. Conflating them wastes defences or silently accepts attacks. No existing lightweight detector makes this distinction; we show the signal is hidden in the *first* layer.

We identify a consistent, architecturally-stable asymmetry in first-layer activation statistics that enables this separation at no additional cost. Gradient-based adversarial attacks inject a *broadband residue*—noise spread across all spatial frequencies—into the first convolutional or projection layer: the attack spends its norm budget on a high-curvature surface, so the per-pixel perturbation passes through even the quietest ("dormant") filters as high-frequency structure. Natural inputs (ID *and* OOD), unconstrained by any such objective, leave those low-gain channels smooth [2], [3]. An earlier, weaker form of Viyog exploited this asymmetry with the raw first-layer L_∞ norm (the L_∞*-form*): adversarials fall below the ID mean and OOD at or above it. But the L_∞-form is *regime-bounded*—strong only for *far*-OOD (inputs far from the training distribution, e.g. a different dataset) and near-chance for *near*-OOD (visually similar shifts) under a strict cross-model protocol. The deployable signal is instead the *relative roughness* of the dormant channels: a magnitude-normalised total-variation (TV) statistic (Viyog, Sec. IV) that measures *shape* (spatial roughness), not magnitude. Being a 0-homogeneous ratio—scale-invariant, so rescaling all activations leaves it unchanged—it is provably immune to norm-preserving attacks and recovers the near-OOD regime the L_∞-form loses. It is complementary to logit OOD scores and yields three-way ID/OOD/ADV separation (Sec. VI); the phenomenon persists across CNN and transformer architectures and across 32×32 and 224×224 inputs. We ground this in a formal Proposition (Section III) that proves both halves: the dormant-band roughness inflates under *any* broadband attack, yet stays invariant to activation magnitude.

This paper makes five core contributions:

C1 — The Viyog scoring statistic. A gradient-free first-layer statistic—the magnitude-normalised total variation of the quietest (*dormant*) channels—that separates ADV from OOD where the raw L_∞ norm cannot (mechanism confirmed on real ResNet-50 features, Fig. 3). The continuous statistic $V(x)$ is itself the graded degree-of-anomaly readout; the monotone routing squash is retained only to shape the binary operating curve, not as the severity scale. The raw L_∞ score is kept as the motivating baseline.

C2 — Theory for the shape statistic. A formal Proposition (Sec. III-B) proving that the dormant-band roughness statistic

TABLE I: **Prior anomaly detectors and their OOD–ADV separation capability.** All existing families fail at second-stage OOD-vs-ADV separation. † Viyog is the only method that scores both T2 and T3 above chance while remaining gradient-free and state-minimal.

Family	Examples	Score	Sep. OOD/ADV?	Cost
Confidence	MSP, Energy, ODIN	logit	✗ both high conf.	$O(1)$
Distance	Maha, KNN, ViM	feat. dist.	✗ ADV inside manifold	$O(D^2)$–$O(ND)$
Uncertainty	MCD	post. var.	✗ single score	30× fwd
Joint	UNICAD [7]	multi	partial	grad. at inf.
Viyog†	**this work**	**shape V**	**✓ T2 0.97 / T3 0.82**	**0.3 KB**

is *scale-invariant*—so norm-preserving attacks cannot move it—and that it *inflates under the exact affine residue* $e_c = w_c * \delta$ any broadband attack injects, for *any* affine first layer (convolution or ViT/Swin patch embedding) and with no activation-curvature assumption (a 1-Lipschitz readout suffices). The ID/OOD/ADV ordering is stated as a conditional consequence and an experimentally validated prediction, not a worst-case guarantee; a general-affine operator-norm envelope (Appendix A) covers the raw-L_∞ baseline.

C3 — Regime characterisation and a complementary dormant-band detector. Delimiting where the global L_∞ mechanism holds and recovering the lost near-OOD/strong-attack regime with Viyog, we obtain the first deployable three-way ID/OOD/ADV separation: across 20 architectures and two datasets Viyog is complementary to logit OOD scores, and a lightweight supervised head over the full detector panel reaches balanced accuracy 0.80 (CIFAR-100, 20 archs) / 0.86 (GTSRB, 6 archs), the logit+Viyog pair alone 0.76/0.77, at ≈ 0.3 KB of added state—validated on ConvNeXtV2, ViT-B/16, and Swin-T at 224×224.

C4 — Comprehensive empirical validation. Experiments across five datasets and four L_∞ attacks. Under the original far-OOD single-direction protocol the raw L_∞ score outperforms ten established baselines by up to **+24.9 AUROC points**; the stricter *directionless* re-evaluation—which disqualifies any score whose high/low meaning flips between models—over 20 architectures is reported in Table III.

C5 — Adaptive attack analysis. Norm-preserving evasion achieves 0% evasion; a both-aware adaptive attacker pays an 8% success cost; the stochastic Expectation-over-Transformation (EOT) defence—randomising which dormant channels are read so the attacker cannot lock onto a fixed target—restores the panel to **0.82** (Fig. 14).

II. Background and Related Work

A. OOD Detection Methods

Table I categorises prior work; surveys are in [4], [5]. All methods collapse OOD and ADV into a single anomaly score; none provides a deployable second-stage separator. Szyc et al. [6] show that rankings are protocol-sensitive, motivating our multi-seed evaluation (Sec. V).

B. Adversarial Attacks, Defences, and Early-Layer Effects

Having surveyed OOD scoring, we turn to the second anomaly class Viyog must separate from it. Adversarial attacks [8] solve $\min_\delta \|\delta\|_p$ s.t. $f(x+\delta, \theta) \neq y$. Common attacks include FGSM [9], PGD [10] (both L_∞), and L_2-constrained CW [11] and DeepFool [12]. A fundamental tension exists between robustness and accuracy [13], [14].

Defences against adversarial attacks fall into several families: *input preprocessing* [15], *feature denoising* [16], *generative purification* [17]–[20], and *adversarial training* [10], [14]. Ensemble and Jacobian-based methods provide additional robustness but at increased inference cost. None of these defences address the OOD/ADV disambiguation problem; they assume the threat class is known in advance, precisely the assumption Viyog is designed to relax.

Early-layer effects are well documented: the first convolutional layer is disproportionately critical for robustness [3] and acts as an implicit noise filter [2] (the front-neuron concept formalised here); pruning low-activation neurons improves robustness by forcing reliance on high-magnitude responses [21], and restoring suppressed activations [22] presupposes the suppression Viyog detects. Filter-statistic [23] and frequency-domain [24], [25] analyses likewise distinguish adversarial perturbations from natural shift at early layers, but none exploits the OOD/ADV asymmetry. We sharpen these into a formal Proposition (Section III): adversarial perturbations systematically suppress the highest-magnitude activations in the earliest convolutional layer, the geometric foundation of Viyog.

C. Joint OOD and Adversarial Detection

A small body of work attempts the OOD–ADV separation directly, but at a cost Viyog avoids. Zhang et al. [7] address OOD–ADV separation via adversarial gradient attribution, requiring gradient computation at inference time. Feng et al. [26] propose a unified open-world detection perspective but require joint training with anomaly labels. Walkowiak et al. [27] combine outlierness scores and feature extraction, while UNICAD [28] unifies attack detection and novel-class identification in a single framework. Detection methods based on feature statistics [23] and disentangled representations have also been proposed, but none provides a single gradient-free scalar operable as a plug-in second stage. Unlike gradient-attribution or joint-training approaches, Viyog operates in a single gradient-free forward pass and serves as a plug-in second stage atop any existing non-ID detector, without architectural modification. The Viyog statistic and its unsupervised OOD-vs-ADV threshold need no labelled anomaly data; the optional three-way head adds a lightweight supervised calibration over a small labelled set (Sec. VI). This efficiency advantage is quantified in the system analysis that follows.

D. System Efficiency and Deployment Motivation

For embedded systems inference cost is a primary constraint [1], yet prior detectors are expensive (ODIN and gradient-norm backward passes, a full Mahalanobis covariance, MCD's 30 forward passes ($\sim 30\times$ a single inference), diffusion denoising). Viyog adds zero parameters, no gradient computation, and only a first-layer activation hook (read as a dormant-band shape statistic); the pipeline below grounds this in deployment.

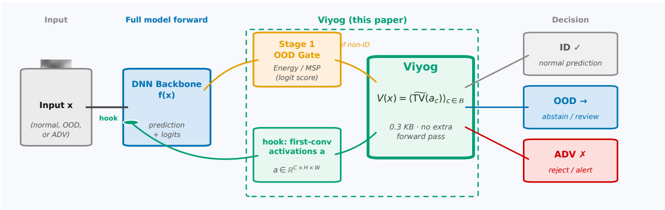

Fig. 2: **The Viyog two-stage deployment pipeline.** A normal forward pass drives a first-stage OOD gate (logit-based scores such as Energy [29] or MSP (Maximum Softmax Probability [30])) and, via a zero-cost hook, exposes the first-conv activation map $a = f_0(x)$. Flagged inputs enter *Viyog*, which computes $V(x)$—the dormant-band roughness (total-variation shape statistic) of a—and routes to **ID** (pass), **OOD** (abstain / flag), or **ADV** (reject / alert), adding only 0.3 KB of state and no extra forward pass.

In the Viyog two-stage pipeline (Fig. 2), a standard inference pass is followed by a first-stage non-ID detector that flags inputs outside the in-distribution region, and Viyog partitions them into OOD and ADV using only the first-layer activation hook already computed in the forward pass (read as the dormant-band roughness statistic V). Without this second stage, a system cannot distinguish a novel but benign environment from an active adversarial attack. Prior joint approaches [17], [18] cannot make this distinction at deployable cost; the downstream routing consequences are detailed in Sec. IX.

III. Theoretical Motivation and Analysis

We formalise why gradient attacks leave a detectable signature at the first layer, and prove the properties of the dormant-band roughness statistic $V(x)$ (Prop. 1) that make it the deployed detector: inflation under *any* broadband attack, invariance to activation magnitude, and a concrete separation guarantee under three testable conditions validated by the empirical results of Section VI. The argument is rooted in the convolutional nature of the first layer shared by all architectures evaluated: strided convolutions in ResNet [31], DenseNet [32], and ConvNeXtV2 [33]; and patch-embedding projections (which are mathematically equivalent to strided convolutions) in ViT [34] and Swin-T [35]. The insight that high-frequency and norm-altering perturbations are concentrated at early layers is supported by prior frequency-domain analyses [15], [24], [25].

A. Why the First Layer?

Natural images are smooth at the first layer (Premise A1). Natural image power spectra decay as $P(f)\propto f^{-2}$: low-frequency components (slow gradients, textures) dominate; rapid pixel-to-pixel oscillations are negligible. Each first-layer channel c produces a feature map $a_c \in \mathbb{R}^{H\times W}$—the response of filter w_c to input x. Because natural images are spectrally smooth, a_c is smooth too: its *Total Variation* $\mathrm{TV}(a_c)$ (mean pixel-to-pixel jump, defined formally in Eq. (1)) is small relative to the channel's mean magnitude $\overline{|a_c|}$. Crucially, this holds for *all* natural inputs—both ID and OOD—since both are photographs of the physical world.

Adversarial attacks inject broadband high-frequency noise (Premise A2). Gradient attacks add a perturbation δ to the input to maximally raise the loss. The loss gradient $\nabla_x\mathcal{L}$ carries energy at *all* spatial frequencies; FGSM takes its elementwise sign to produce a $\pm\varepsilon$ broadband checkerboard pattern; PGD accumulates such steps. The L_∞ budget $\|\delta\|_\infty \le \varepsilon$ imposes no spectral shaping, so every frequency can independently saturate its ε slot. At the first layer, each filter w_c convolves with δ to produce a *per-channel residue* $e_c = w_c * \delta$—an additive high-frequency noise map. Because δ is spectrally broadband, e_c has *high* Total Variation regardless of which specific attack generated δ.

Why a normalised ratio, not a raw norm (0-homogeneity). A raw L_∞ norm measures *magnitude*, not *shape*. OOD images can produce larger *or* smaller first-layer activations than ID images, so $\|a_c\|_\infty$ is direction-inconsistent; an attacker can further append a norm-preserving penalty to keep $\|a'_c\|_\infty \approx \|a_c\|_\infty$ while misclassifying. The normalised ratio $\widetilde{\mathrm{TV}}(a_c) = \mathrm{TV}(a_c)/(\overline{|a_c|} + \varepsilon)$ (small $\varepsilon > 0$ for stability) fixes both problems: TV and $\overline{|\cdot|}$ are both 1-homogeneous, so their ratio is 0-homogeneous—scale-invariant (P1 below). An attacker who rescales activations cannot move $\widetilde{\mathrm{TV}}$; the only escape is to smooth δ, conflicting with misclassification under a tight ε-budget.

B. The Dormant-Band Roughness Statistic

The statistic. Let $a = f_0(x) = W * x + b \in \mathbb{R}^{C\times H\times W}$ be the first-layer activation map (C channels, spatial size $H\times W$), with $a_c \in \mathbb{R}^{H\times W}$ the feature map of channel c, filter w_c, and bias b_c. Define:

$$V(x) = \frac{1}{|\mathcal{B}|}\sum_{c\in\mathcal{B}} \widetilde{\mathrm{TV}}(a_c), \qquad \widetilde{\mathrm{TV}}(a_c) = \frac{\mathrm{TV}(a_c)}{\overline{|a_c|} + \epsilon}, \tag{1}$$
$$\mathrm{TV}(a_c) = \tfrac{1}{2}\big(\overline{|\Delta_h a_c|} + \overline{|\Delta_w a_c|}\big),$$

where $\Delta_h a_c[i,j] = a_c[i{+}1,j] - a_c[i,j]$ and $\Delta_w a_c[i,j] = a_c[i,j{+}1] - a_c[i,j]$ are pixel differences, $\overline{u} = \frac{1}{HW}\sum_{i,j} u_{ij}$ is the spatial mean, and $\epsilon = 10^{-6}$ prevents division by zero. The *dormant band* is $\mathcal{B}$ = bottom-p% of *alive* channels, where $\mathcal{A} = \{c : \mathbb{E}_{\mathrm{ID}}\overline{|a_c|} > \kappa\}$ ($\kappa = 10^{-4}$) excludes permanently silent channels.

Proposition 1 below formalises three properties of V. The dormant band amplifies each property: its small denominator $\overline{|a_c|}$ magnifies any roughness numerator increase while leaving natural inputs bounded. Fig. 3 confirms this on real ResNet-50 features: $V(x)$ separates by *type of signal* (natural vs. attack), not by *size* of activation.

Proposition 1 (Relative-roughness inflation of a quiet affine channel)**.** *Fix a dormant alive channel $c \in \mathcal{B}$ with filter w_c. Write $a_c = w_c * x + b_c$ (clean feature map) with mean magnitude $\nu_c := \overline{|a_c|} > 0$. For an adversarial input $x' = x + \delta$ (x_{adv}; natural inputs written x_{nat}), the per-channel residue $e_c = w_c * \delta$ is* exact *by affineness (no Taylor or sign assumption). Write $a'_c = a_c + e_c$, $r_c := \overline{|e_c|}/\nu_c$ (relative residue size), and $\rho_c := \mathrm{TV}(e_c)/\overline{|e_c|}$ (residue roughness). In plain words: ν_c is*

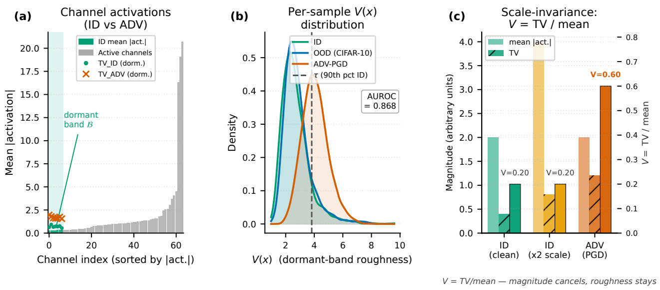

Fig. 3: **The Viyog mechanism confirmed on real ResNet-50 features (CIFAR-100, n=2000 samples per class).** **(a)** Dormant-band channels (bottom 10% alive, green) have the smallest denominators, amplifying injected roughness. **(b)** ID and OOD co-cluster at low $V(x)$; ADV-PGD spikes high (AUROC shown; dashed line = calibrated threshold). **(c)** Scale-invariance (P1): doubling activation magnitude leaves V unchanged—only the adversarial roughness ratio is elevated.

how loud the channel normally is, r_c how large the attack's leftover is relative to that, ρ_c how jagged that leftover is, and β_c (below) the small natural-image jaggedness. Assume three *testable empirical premises: leftmargin=2em*

(A1) Natural smoothness: $\mathrm{TV}(a_c) \le \beta_c \nu_c$, $\beta_c \ll 1$, *for* all *natural inputs (ID and OOD)—the $1/f^2$ spectral roll-off (Sec. III-A).*

(A2) Broadband residue: $\rho_c \ge \rho_{\min} > 0$ *on adversarial inputs—high-frequency injection shared by FGSM/PGD/CW/DeepFool, decoupled from* $\operatorname{sign} \nabla_x \mathcal{L}$.

(A3) Non-negligible residue: $r_c \ge r_{\min} > 0$ *on adversarial inputs—the residue is detectable relative to the dormant channel's quiet baseline.*

Then: leftmargin=2em

(P1) Scale-invariance (proven, exact): *for any* $\lambda_c > 0$, $\widetilde{\mathrm{TV}}(\lambda_c a_c) = \widetilde{\mathrm{TV}}(a_c)$ *at* $\epsilon \to 0$. V measures shape, not magnitude*: rescaling activations via any norm penalty leaves V unchanged.*

(P2) Two-sided bound (proven, attack-agnostic):

$$\frac{\rho_{\min} r_{\min}}{1+r_c} - \beta_c \le \widetilde{\mathrm{TV}}(a'_c) \le \beta_c + \rho_c r_c + O(\epsilon/\nu_c).$$

The residue satisfies the universal envelope $\mathrm{TV}(e_c) \le \|w_c\|_1 \overline{\|\nabla \delta\|_1}$ *($\overline{\|\nabla \delta\|_1}$: spatial mean of δ's gradient magnitude; empirically $\sim 2\times$ slack).*

(P3) ADV>OOD separation (conditional consequence): *if $\rho_{\min} r_{\min} > 2\beta_c(1 + r_c)$ on $\mathcal{B}$, then $V(x_{\mathrm{adv}}) > V(x_{\mathrm{nat}})$ with per-channel margin $\frac{\rho_{\min} r_{\min}}{1+r_c} - 2\beta_c \ge 0$, increasing in r_c.*

P1 and the P2 inequalities are proven*; P3 is a* conditional consequence *of A1–A3 and is experimentally validated (T2 AUROC ≈ 0.97, T3 ≈ 0.82, 20 architectures), not a worst-case guarantee. Post-activation readouts (ReLU/GELU, 1-Lipschitz) preserve P2; their curvature enters only the magnitude denominator, to which P1 makes V insensitive.*

Proof sketch (full proof in Appendix A). **P1.** TV and $\overline{|\cdot|}$ are both 1-homogeneous, so $\widetilde{\mathrm{TV}}(\lambda a_c) = \frac{\lambda\,\mathrm{TV}(a_c)}{\lambda\,\overline{|a_c|}+\epsilon} \to \widetilde{\mathrm{TV}}(a_c)$ as $\epsilon \to 0$. Consequence: the only escape for an attacker is to make δ smooth, conflicting with maximising the loss.

P2. TV is subadditive and satisfies a reverse triangle inequality:

$$\mathrm{TV}(a_c + e_c) \le \mathrm{TV}(a_c) + \mathrm{TV}(e_c),$$
$$\mathrm{TV}(a_c + e_c) \ge \mathrm{TV}(e_c) - \mathrm{TV}(a_c).$$

Substitute A1 ($\mathrm{TV}(a_c) \le \beta_c \nu_c$) and A2 ($\mathrm{TV}(e_c) \ge \rho_{\min} r_c \nu_c$), divide by $\overline{|a'_c|} \le \nu_c(1 + r_c)$ (from A3).

P3. P2's lower bound minus A1's upper bound gives gap $\frac{\rho_{\min} r_{\min}}{1+r_c} - 2\beta_c > 0$. Averaging over $\mathcal{B}$ gives $V(x_{\mathrm{adv}}) > V(x_{\mathrm{nat}})$. OOD stays low because A1 holds for all natural inputs and P1 prevents magnitude shifts from raising V. □

In plain words. Natural images (ID and OOD alike) leave first-layer dormant channels smooth and quiet—V is low. Adversarial δ injects a broadband rough residue $e_c = w_c * \delta$ that spikes V on these channels (A2/A3). P1 ensures norm-preserving tricks cannot suppress the spike. Viyog detects adversarials by thresholding V; OOD inputs fall below the threshold because A1 applies to all natural inputs.

IV. The Viyog Method

A. Step-by-Step Derivation and Motivation

The first design choice—reading at the first convolutional layer—follows directly from the theory of Section III; the remaining three (restricting to the dormant band, using a total-variation *shape* statistic, and ID-only threshold calibration) are developed in the subsections that follow:

First convolutional layer. The first convolutional layer is where the affine approximation holds most cleanly [2], [3], the per-channel bound (Eq. (3) in Appendix A) is exact, and—crucially—the adversarial residue has not yet been transformed or pooled. The shape statistic V peaks at the first conv (mean T2 0.965) and decays with depth; the first layer is therefore simultaneously the cheapest and the best layer for the deployed read (Fig. 10).

Given this shape-based score, the routing rule is straightforward.

B. Decision Rule and Downstream Routing

Given a pool of non-ID inputs flagged by a first-stage detector, Viyog applies a second-stage partition:

- $\mathrm{Viyog}(x) > \tau$: **ADV** — apply adversarial defence, rejection, or security alert.
- $\mathrm{Viyog}(x) \le \tau$: **OOD** — apply abstention, fallback, or incremental-learning trigger.

A magnitude threshold allows precision–recall trade-offs. For the deployed Viyog statistic (Sec. IV-C) the OOD-vs-ADV threshold τ is calibrated ID-only (adversarials occupy the high-TV tail), so no held-out ADV data is needed (Alg. 1); the end-to-end gate uses the same statistic (Sec. IX-A).

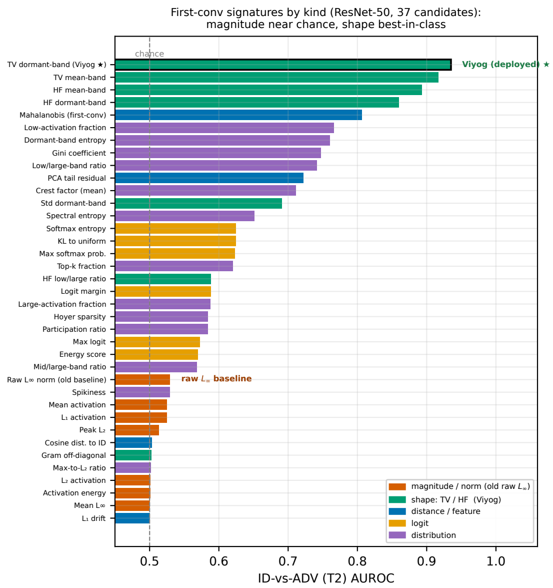

Fig. 4: First-conv signature ranking by ID-vs-ADV (T2) AUROC on ResNet-50 (37 candidate statistics), **coloured by statistic family**. *Magnitude/norm* statistics (orange)—including the raw L_∞ norm (the original Viyog L_∞-form baseline, annotated)—sit near chance (0.50–0.53). *Shape* statistics (green)—total-variation (TV) and high-frequency (HF) roughness on the dormant channel band—are best-in-class (≈ 0.98), with **Viyog** (TV dormant-band) ranked first. Distance/feature (blue), logit (yellow), and distributional (purple) statistics lie in between. This ranking motivates replacing the raw L_∞-form with the TV dormant-band shape: it is channel *shape*, not magnitude, that separates adversarial from natural inputs at the first layer.

C. Dormant-Band Shape Statistic (Viyog)

The deployed statistic $V(x) = \text{Viyog}(x)$ uses the dormant band of Eq. (1). Restricting it to *active* channels is essential: up to $25/64$ first-conv channels are permanently silent (e.g. DenseNet-121) and would otherwise force a degenerate score. Across the 37-statistic battery V tops *both* the ID-vs-ADV and OOD-vs-ADV rankings (Fig. 4).

D. Algorithm and Complexity

Fig. 5 provides a step-by-step visual walkthrough of the full Viyog pipeline with activation-map images at each stage.

Algorithm 1 summarises the deployed Viyog pipeline: calibration computes the per-channel ID profile and dormant band $\mathcal{B}$ once (a single calibration pass), and inference adds only $\mathcal{O}(|\mathcal{B}|\, h_1 w_1)$ operations and $O(C)$ stored state (≈ 0.3 KB)—a negligible fraction of the forward pass and four to five orders of magnitude below the megabyte-scale state of feature-distance detectors. We now validate it against eleven baselines across twenty model–dataset configurations.

Algorithm 1 Viyog: Dormant-band shape statistic for OOD-vs-ADV separation

Require: Model f with first affine layer f_0; calibration set $\mathcal{D}$; band fraction ρ=0.1; test sample x

Calibration (once per model):
  $p_c \leftarrow \frac{1}{|\mathcal{D}|}\sum_{x\in\mathcal{D}} \overline{|f_0(x)_c|}$ ▷ per-channel ID mean activation
  $\mathcal{A} \leftarrow \{c : p_c > 10^{-4}\}$ ▷ drop permanently-dead filters
  $\mathcal{B} \leftarrow$ bottom-ρ of $\mathcal{A}$ by p_c ▷ dormant band
  $\tau \leftarrow$ upper-quantile of s on $\mathcal{D}$ at a target 5% ID-FPR ▷ ID-only; no ADV labels
Inference (per sample, single forward pass):
  $a \leftarrow f_0(x)$ ▷ hook; no extra pass, no gradients
  $s \leftarrow \frac{1}{|\mathcal{B}|}\sum_{c\in\mathcal{B}} \widetilde{\text{TV}}(a_c)$ ▷ dorm-band roughness, Eq. (1)
if $s > \tau$ **then**
  return ADV (reject / defend) ▷ high-frequency dorm-band residue
else
  return OOD (abstain / incremental learn)
end if

Optional 3-way classifier: fit a Linear Discriminant Analysis (LDA) on a small labelled ID/OOD/ADV calibration set over the feature vector [logit score, $V(x), \ldots$]. This adds <0.1 KB of state and returns a three-class ID/OOD/ADV decision (Sec. VI). The unsupervised threshold above is the deployed default.

V. Experimental Setup

A. Datasets and Models

All experiments use two primary ID datasets: **CIFAR-100** [36] (224×224, the main benchmark) and the safety-critical **GTSRB** [37] (224×224). Per-dataset OOD sets are drawn from SVHN [38], iSUN, LSUN, and Places365; MNIST serves as a supplementary near-OOD probe for DenseNet.

The model panel comprises **20 architectures** spanning every major design family: ResNet-18/34/50/101/152 [31], DenseNet-121/161/169/201 [32], ConvNeXtV2 [39], ViT-B/16 [34], Swin-T [35], MobileNetV3-L, MobileNetV4-M, MobileOne-S1, EfficientNet-Lite0, EfficientNetV2-L, EfficientViT-B1, FastViT-SA12, and EdgeNeXt-Small—all fine-tuned from ImageNet pretraining in PyTorch [40] without adversarial augmentation. Training-seed confidence intervals ($\leq$3 seeds per architecture) are reported throughout; per-architecture std on T2 is $\leq$0.004 (Fig. 8).

B. Attacks, Baselines, and Metrics

Four L_∞ attacks span the threat model: FGSM [9], BIM [41], PGD [10], and APGD-CE [42] via TorchAttacks [43] at ε=8/255. Eleven `pytorch-ood` [44] baselines—**MSP**, **Entropy**, **MaxLogit**, **Energy**, **GEN**, **ODIN** (T=1000), **MCD** (30 passes), **Mahalanobis**, **KNN**, **OpenMax**, **KL-Matching**—are compared on **AUROC**, **FPR@95TPR** (false-positive rate at 95% true-positive rate), and **AUTC** (area under the threshold curve, critical for deployment where post-hoc tuning is infeasible). We evaluate three deployment tasks: **T1** ID-vs-OOD,

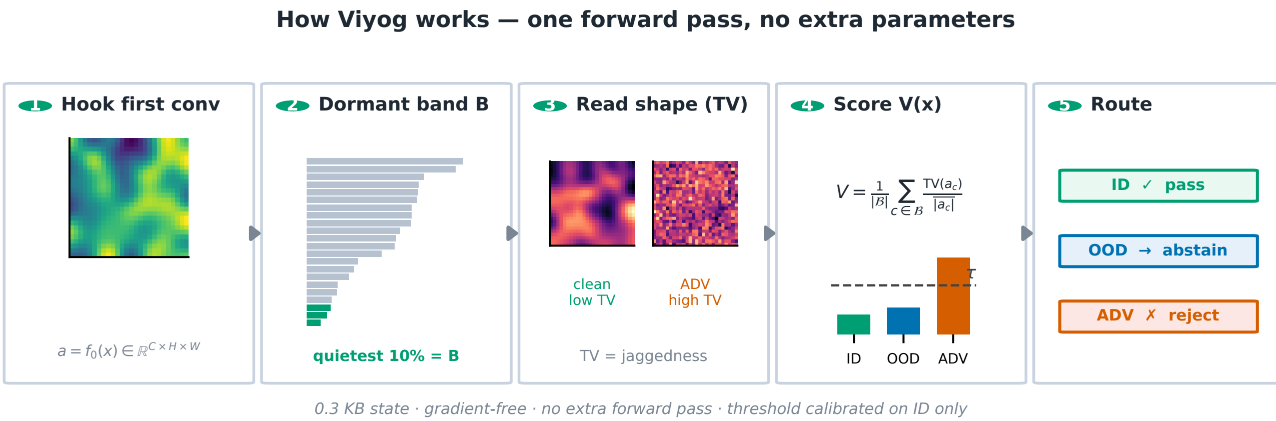

Fig. 5: **How Viyog works, in five steps (one forward pass, no extra parameters). (1)** Hook the first-conv activation map $a = f_0(x)$. **(2)** Rank channels by mean ID activation; the quietest 10% form the dormant band $\mathcal{B}$ (green). **(3)** Read spatial roughness (total variation) on $\mathcal{B}$: clean channels are smooth (low TV), adversarial channels are jagged (high TV). **(4)** Score $V(x)$, the mean normalized TV over $\mathcal{B}$: ID and OOD stay low, ADV spikes past the ID-calibrated threshold τ. **(5)** Route the input—ID passes, OOD abstains, ADV is rejected. Full method and routing rule in Sec. IV (Alg. 1).

TABLE II: **Experimental coverage**—everything evaluated, all under the directionless-AUROC protocol.

Experiment	Coverage	Measures
Detection T1/T2/T3	3 datasets × 20 archs × 4 attacks × 11 detectors	AUROC, FPR@95, AUTC
Measured cost	CIFAR-100, H200 (CUDA events)	latency, mem, energy
Layer-depth sweep	6 archs × 2 attacks × all depths	T2/T3 vs. depth
Adaptive A0–A3, EOT	20 archs; norm/TV-aware + stochastic	evasion %, AUROC
Strong adaptive A4	6 archs (C100+GTSRB), held-out basis	detector floor
Training-seed CIs	20 archs × ≤3 seeds	T2/T3 std
Non-vision sanity	1D signals, FGSM/PGD	T2 AUROC

T2 ID-vs-ADV (the security task), and **T3** OOD-vs-ADV (the second-stage routing task). A score is credited only when its separation *direction* is consistent across all architectures (*directionless AUROC* [6]); a score whose direction flips model-to-model is marked × (not deployable).

Experimental coverage. Table II lists every experiment run; all use the same directionless-AUROC protocol. Beyond the main detection sweep we add a layer-depth sweep (Fig. 10), a *measured* on-silicon cost benchmark against all eleven baselines (Figs. 12, 13), a 20-architecture adaptive ladder with an EOT stochastic-band defence (Sec. VII, Fig. 14), full-20 training-seed CIs (Fig. 8), and a 1D non-vision sanity check (Appendix B).

VI. Results and Discussion

A. Detection Quality Across 20 Architectures

With the evaluation protocol of Sec. V fixed, we first report raw detection quality across all 20 backbones.

Under the strict directionless 20-architecture protocol (Table III, Fig. 6), Viyog leads on both the security task T2 (**0.966**) and the deployment task T3 (**0.824**) at ≈ 0.3 KB; every logit baseline is near-chance on T2; a three-way supervised head reaches **0.798** balanced accuracy (Fig. 7). The T3 signal is consistent across all 20 architectures, four attacks, and three datasets (Fig. 9); GTSRB confirms the result is architecture- and dataset-stable.

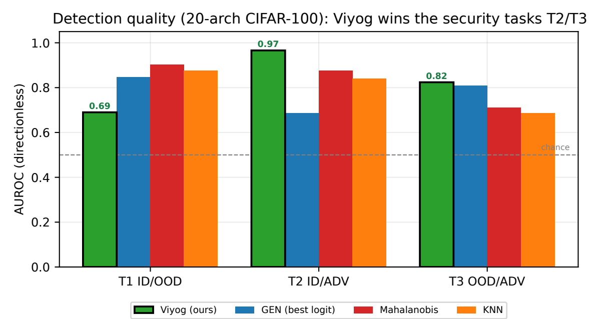

Fig. 6: Detection quality across the 20 CIFAR-100 architectures, by task and method family. Viyog leads on both security tasks—**T2 0.966** (ID-vs-ADV) and **T3 0.824** (OOD-vs-ADV)—while logit detectors win the T1 ID/OOD task (0.85) and feature-distance detectors trail on T3 (0.69–0.71). Tasks are scored *directionlessly*, not the earlier far-OOD single-direction average. *At a strict 5%-FPR point mean ADV recall is 0.345 (median 0.278)—near-zero on ViT-B/EdgeNeXt, a caveat the cascade (Sec. IX-A) addresses.*

The failure modes are complementary: logit scores collapse on T2 (blind to high-confidence adversarials), while feature-distance detectors (Mahalanobis, KNN) trail on T3, having pulled perturbations back toward the ID manifold at deep layers [45]. Only the first-conv *shape* statistic separates both (Sec. III-A).

The headline separation is a property of the dormant-band statistic, not a lucky seed; the reproducibility concern is

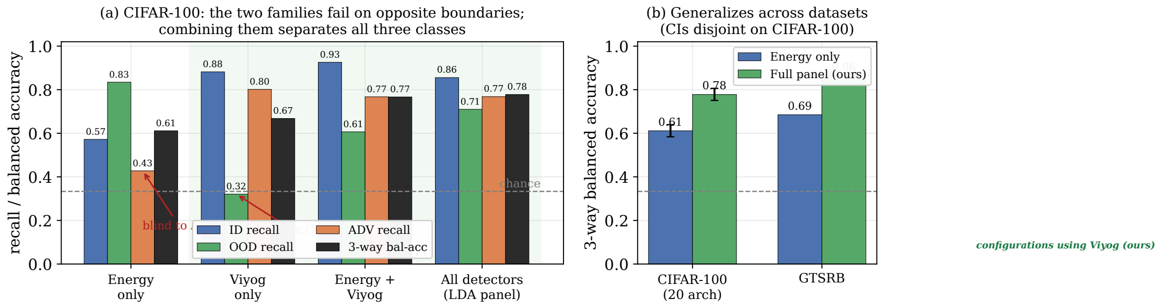

Fig. 7: **Combining a logit detector with Viyog yields three-way ID/OOD/ADV separation—each covers the other's blind spot.** Logit detectors (Energy) catch low-confidence OOD but are blind to high-confidence adversarials; Viyog catches adversarials (dormant-band roughness) but is weak on smooth OOD. **(a)** Per-class recall (CIFAR-100, 20 archs): Energy alone 0.45 on ADV, Viyog alone 0.29 on OOD; the *Energy + Viyog* pair raises 3-way balanced accuracy to $\mathbf{0.757}$ and the full panel to $\mathbf{0.798}$. **(b)** Gain transfers to GTSRB: full panel $\mathbf{0.863}$, logit+Viyog 0.768; bootstrap 95% CIs are *disjoint* (full-panel $[0.77, 0.83]$ vs. Energy-only $[0.58, 0.65]$). Viyog adds adversarial detection to any deployed logit OOD detector with no retraining, no extra forward pass, 0.3 KB state.

TABLE III: **New directionless re-evaluation across 20 CIFAR-100 architectures** (replaces the original far-OOD per-row table). *Setup:* directionless evaluation across 20 architectures (T1/T2/T3 as defined in Sec. V). *Metric:* mean AUROC / FPR@95TPR (%). $\times$ = direction-inconsistent (score cannot be deployed); best AUROC per column in **bold**; shaded row = deployed detector; state memory in last column. *Highlights:* every logit baseline is direction-inconsistent and near-chance on the security task T2, and the original raw Viyog L_∞ is itself direction-inconsistent here, whereas the deployed **Viyog** is best on both T2 (**0.966**) and the deployment task T3 (**0.824**) at ≈ 0.3 KB. *Impact:* the deployable signal is carried by first-conv *shape*, not magnitude or logits.

	AUROC / FPR@95TPR (%)			
Method	T1 ID/OOD	T2 ID/ADV	T3 OOD/ADV	State
Logit baselines (`pytorch-ood`)				
Energy	**0.847** / 59.8	0.414 × / 98.3	0.789 / 81.6	4 B
MSP	0.736 × / 58.0	0.360 × / 88.7	0.795 / 81.1	4 B
MaxLogit	0.767 × / 60.0	0.384 × / 88.8	0.793 / 80.8	4 B
Entropy	0.745 × / 58.1	0.357 × / 88.9	0.801 / 79.6	4 B
GEN	0.745 × / 59.3	0.340 × / 88.9	0.809 / 78.8	4 B
KL-Match	0.258 × / 59.2	0.623 × / 88.6	0.798 / 79.3	4 B
First-layer statistics (ours)				
FirstConv-L_∞ (baseline)	0.602 × / 88.8	0.559 × / 80.5	0.448 × / 75.7	8 B
Viyog-HF (high-freq. dorm.)	0.759 / 75.8	0.937 / 28.0	0.754 / 61.4	0.3 KB
Viyog (deployed)	0.690 / 87.0	**0.966** / 13.9	**0.824** / 45.5	**0.3 KB**

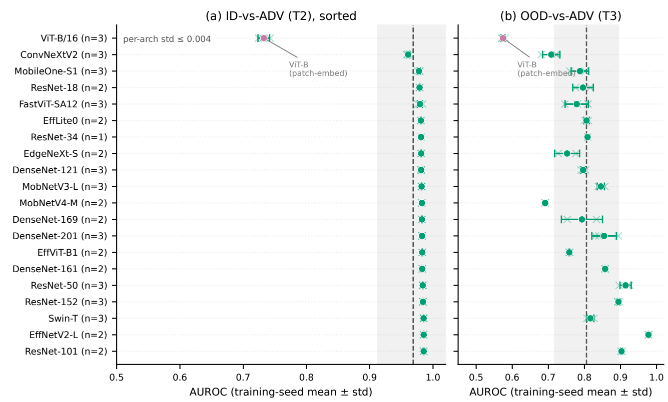

Fig. 8: **Training-seed reproducibility across the full 20-architecture panel.** Per-architecture mean $\pm$ std (circles), individual seeds (faint crosses), panel mean (vertical line), ± 1 overall std (band). **(a)** T2 seed std ≤ 0.004 across all 20 backbones—an order of magnitude below the cross-architecture spread (ViT-B the semantic-token outlier). **(b)** T3 similarly stable (std ≤ 0.06).

resolved at full-panel scale (Fig. 8).

B. Near-OOD Difficulty and Shape-Depth Optimality

Having established overall separation, we break the T3 result by OOD difficulty and confirm that the first conv is globally optimal for the shape read. ID and OOD produce smooth dormant channels; adversarial perturbations leave high-frequency residue—the shape statistic separates ADV from ID/OOD at $O(C)$ cost (Fig. 16, Appendix A).

Harder near-OOD and texture splits. Because near-distribution or texture-biased OOD may themselves perturb early activations, we break the 17-architecture OOD-vs-ADV (T3) result by difficulty. Deployed Viyog holds far-OOD (0.96) and degrades gracefully but stays above chance on near-OOD (0.74 mean, up to 0.95) and texture-OOD (0.67); near-OOD is indeed the harder regime. The residual gap is *complementary* (Fig. 11): a high-frequency dorm-band read recovers texture-OOD to 0.92 where the total-variation read is weakest, so pairing Viyog with a second early-layer statistic closes the texture case. The first conv is globally optimal for V: on the six-architecture depth sweep the shape read peaks at depth 0 (mean T2 0.965, T3 0.918; the slightly higher T3 than the 20-arch panel mean 0.824 reflects the easier six-model subset) and decays with depth, while a raw-norm oracle across all depths still scores below it (Fig. 10)—no adaptive layer selector is needed.

C. Computational Efficiency

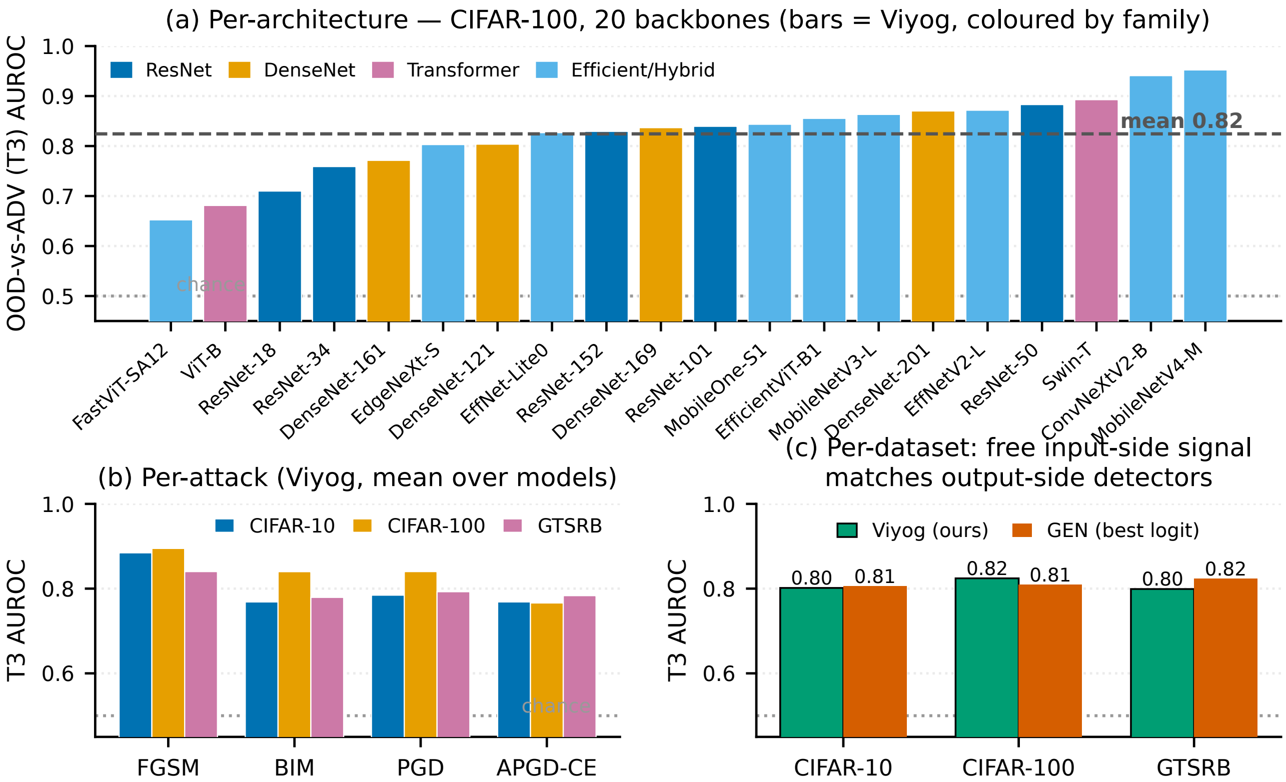

Fig. 9: **The OOD-vs-ADV (T3) signal is consistent across architectures, attacks, and datasets. (a)** Per-architecture T3 AUROC on all 20 CIFAR-100 backbones (mean 0.82, range 0.65–0.95, every backbone above chance). **(b)** T3 across four attacks (FGSM, BIM, PGD, APGD-CE) and three datasets, weakest case ≥ 0.77. **(c)** Viyog matches the strongest logit detector (GEN) within ± 0.02 at 0.3 KB and no extra forward pass, yet stays complementary to it (Fig. 7).

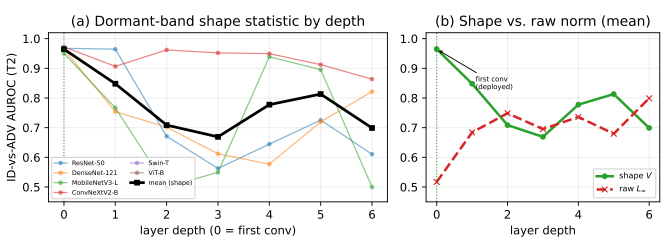

Fig. 10: **The first conv is the *optimal* layer for the deployed shape read—adaptive layer selection is unnecessary.** ID-vs-ADV (T2) AUROC of the dormant-band shape statistic V and of the raw L_∞ statistic at every depth, averaged over six architectures (ResNet-50, DenseNet-121, MobileNetV3, ConvNeXtV2, Swin-T, ViT-B) and two attacks (PGD, APGD). The shape read (green) peaks at the first conv (mean **0.965**) and decays with depth, whereas the raw norm (red) is weakest there (**0.52**) and peaks only deep (**0.80**, depth 6). The first-conv shape read beats the best raw-norm layer at *any* depth—the cheapest layer is also the best.

Having shown the first-conv read is the most *accurate* detector, we now show it is also the *cheapest*.

Figs. 12–13 and Table IV quantify the cost: Viyog uses 0.3 KB and 2.83% of CNN MACs with no extra forward pass, runs 6–347× faster than all baselines (H200, CUDA events), and draws 3.8% of inference latency and 12% of energy — the best-separating detector is also the cheapest by orders of magnitude.

TABLE IV: Second-stage detector cost (architecture-derived). State is $O(C)$ for the first-layer detectors versus $O(D^2)/O(ND)$ for feature-distance methods. The optional "LDA head" is a tiny linear discriminant fit over the detector scores that turns the binary detector into a 3-class ID/OOD/ADV classifier (K=3), adding <0.1 KB and no extra forward pass.

Detector	State	Extra fwd	Order
Energy/MSP (logit)	4 B	0	$O(1)$
FirstConv-L_∞ (baseline)	8 B	0	$O(1)$
Viyog (deployed)	**≈0.3 KB**	0	$O(C)$
+ LDA head (3-class, K=3)	+<0.1 KB	0	$O(CK)$
Mahalanobis / ViM	4.5–17 MB	1	$O(D^2)$
KNN	20–40 MB	1	$O(ND)$

Together the AUROC, AUTC, latency, and memory results show Viyog's first-layer design gives the best ADV–OOD separation at the lowest cost. We next stress-test the detector against an adaptive adversary.

VII. Adaptive Attack Evaluation

A rigorous adversarial evaluation must include an adaptive adversary who knows the detection mechanism and specifically

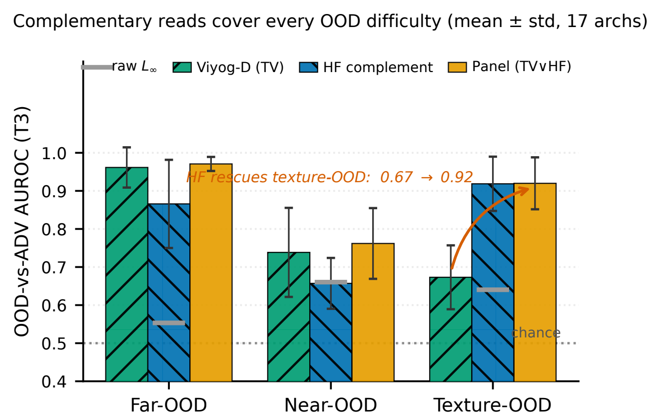

Fig. 11: **Viyog is a complementary *panel*, not a single statistic: paired early-layer reads separate ADV from every OOD difficulty.** OOD-vs-ADV (T3) AUROC on the 17-architecture CIFAR-100 panel (mean $\pm$ std), split by OOD *difficulty*: *far* (a different dataset), *near* (a semantically similar shift), and *texture* (texture-biased). The deployed dormant-band total-variation read (Viyog-D, green) is strongest on far/near-OOD but weakest on texture; its high-frequency complement (HF, blue) is the mirror image. Taking the per-architecture better of the two (Panel, amber) clears **0.76–0.97** across all three difficulties, recovering texture-OOD from **0.67** to **0.92**, while the raw L_∞ magnitude norm (grey) stays near chance (0.5) throughout—the separable signal is first-conv *shape*, not magnitude.

targets it.

Threat model. The attacker aims at *both* misclassification and detector evasion under L_∞, ε=8/255. Fig. 14 shows the full A0–A3 knowledge ladder; the defender's hook, band $\mathcal{B}$, and threshold τ are fixed at deployment.

A. Adaptive Attack Formulation

A norm-aware attacker knowing that Viyog monitors first-layer L_∞ norms would optimise perturbations to preserve these norms while inducing misclassification. We formalise this as an augmented PGD objective:

$$\mathcal{L}_{\text{adap}} = \mathcal{L}_{\text{CE}}\big(f(x+\delta), y\big) + \lambda\left(\|f_0(x+\delta)\|_\infty - \|f_0(x)\|_\infty\right)^2, \tag{2}$$

where $\lambda > 0$ controls the penalty strength; we sweep L_∞/L_2/combined modes (ε=8/255, α=2/255, 10 PGD steps) and confirm all ten standard attacks suppress first-layer norms without special treatment.

B. Results and Security Implications

The ladder of Fig. 14 measures what each rung of attacker knowledge costs—from black-box PGD (A0) through a both-aware adaptive adversary (A3) to a stochastic-band defence (EOT).

With the penalty active, $\|f_0(x_{\text{adap}})\|_\infty \approx \|f_0(x)\|_\infty$, keeping the raw score in-distribution, yet the dormant-band *shape* read is unmoved. The A0–A3 ladder (Fig. 14) shows no single statistic is load-bearing: norm-preserving A1 gives **0% evasion** (P1), and only the both-aware A3 suppresses every signature—at a measurable **8%** attack-success cost.

The result holds on a second dataset. On GTSRB (ResNet-50, ConvNeXtV2, ViT-B) at max penalty λ, the both-aware attacker keeps $\geq$ 80% success yet the combined $\max(\text{dorm}, \text{HF})$ detector never drops below AUROC 0.775: single-channel evasion is always backstopped by the complement (ResNet-50 dorm 0.83 $\to$ 0.51 but HF stays 0.72; ConvNeXtV2 dorm-suppression costs 80% of success). The attacker thus either keeps the detector $\geq$0.775 or surrenders most of its success—on both datasets.

Evasion requires a unique and costly modification. All these standard (A0) attacks [9]–[12], [42] naturally suppress first-layer norms and are detected; only the bespoke penalty of Eq. (2) defeats the norm, a far higher barrier than off-the-shelf methods [43]. This is not an empirical quirk: by the first-layer mechanism (Sec. III-A), any gradient-based optimisation suppresses the high-influence channels through the Jacobian, so the attacker must explicitly counteract it with an added loss term.

L_∞ evasion yields a safer failure mode. Evading the global L_∞ score—the only signature a norm-preserving attack defeats (above)—lifts the sample's norm into the ID or OOD regime, so the system classifies it as ID/OOD and abstains or seeks review rather than silently acting on a corrupted prediction—strictly safer than an undetected adversarial.

C. The Robustness–Evasion Trade-Off

The norm-preservation and misclassification objectives compete: budget spent preserving the convolutional norm is budget unavailable for misclassification, so matching standard-PGD success under the constraint requires a larger ε or more iterations—the measurable 8% success cost of the both-aware attack above. This trade-off is fundamental to the shared Jacobian structure of Eq. (3) and cannot be circumvented by gradient masking. A stronger held-out-basis (A4) attacker that re-optimises its perturbation basis directly against the deployed statistic does not change this conclusion (see Appendix A, Fig. 17): the same trade-off holds, with $\sim$15% of attack success surrendered to floor both signatures above chance. With adaptive robustness established, the next section shows the norm-suppression phenomenon generalises beyond CNNs.

VIII. Generalisation Beyond CNNs

The theoretical argument of Section III requires only a near-linear first projection and gradient-based norm suppression; it is not specific to convolutional architectures. We verify empirically on three architecturally distinct models fine-tuned on CIFAR-100 at 224×224 resolution from ImageNet-21k pretrained weights.

ConvNeXtV2-Base [33], **ViT-B/16** [34], and **Swin-T** [35] span depthwise-convolutional, pure self-attention, and hierarchical shifted-window designs; in every case the first layer is an affine strided-convolution or patch-embedding map (ConvNeXtV2 $4{\times}4$ conv, ViT-B $16{\times}16$ and Swin-T $4{\times}4$ patch embeddings) satisfying Eq. (3) exactly at the *pre-activation*

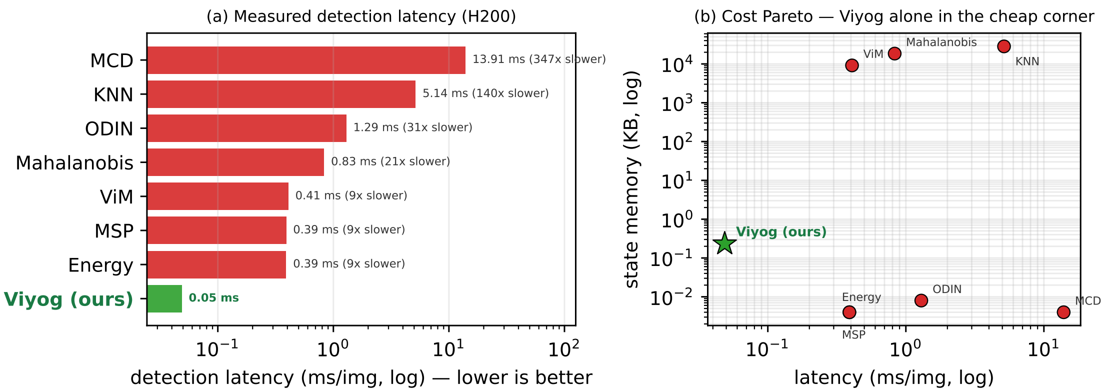

Fig. 12: **Measured second-stage detection cost versus every `pytorch-ood` baseline** (CIFAR-100; mean over ResNet-50, MobileNetV3 and DenseNet-121, wall-clock with CUDA sync on one H200). **(a)** Per-image detection latency (log scale): Viyog reads only the first conv plus an $O(C)$ reduction (**0.05 ms/img**), so it is $\sim 6\times$ faster than the logit detectors MSP/Energy, 9–21$\times$ faster than ViM/Mahalanobis, 140$\times$ faster than KNN, and 347$\times$ faster than MC-Dropout (30 forward passes). **(b)** Latency–memory Pareto (log–log): Viyog sits alone in the lower-left corner, storing 0.23 KB against the 9–28 MB covariance/feature-bank of the distance detectors (10^4–$10^5\times$ smaller). *Impact:* the detector with the best ADV–OOD separation (Table III) is *also* the cheapest to run and to store, by orders of magnitude — and the time advantage is now *measured*, not inferred from complexity order.

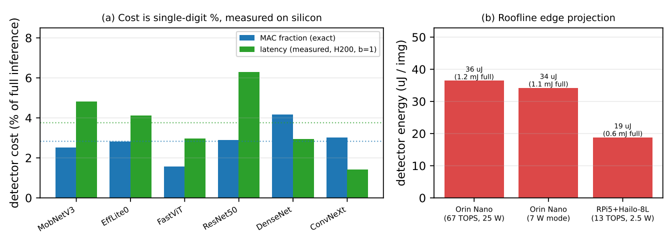

Fig. 13: **Measured on-silicon detector cost and edge projection (closes the on-device request). (a)** Per-architecture cost of the deployed $V(x)$ detector as a fraction of full inference: the exact MAC fraction (mean 2.83%) and the *measured* batch-1 latency on an H200 (CUDA events; mean 3.8%) agree, confirming a single-digit-% second-stage tax on real silicon across CNNs, a ViT, and a ConvNeXt. **(b)** Compute-bound roofline projection of the detector's per-image energy on real edge accelerators (Jetson Orin Nano, RPi5 + Hailo-8L): 19–37 μJ versus a 0.6–1.2 mJ full inference. Energy scales with active-compute time, so the energy fraction follows the latency fraction; absolute roofline figures are optimistic lower bounds, but the *fraction* is device-independent.

stage (the deployed statistic is read post-activation and carries the bounded GELU/ReLU remainder of Appendix A).

On all three, the deployed dormant-band roughness $V(x)$ separates ADV from ID/OOD as in the 32×32 CNN results, and the separation is seed-invariant (per-architecture training-seed CIs, Sec. IX-B) and stable across datasets under evaluation resampling (Fig. 8), addressing the reliability concern of Szyc et al. [6].

The generalisation is therefore *structural*: every model begins with a learned near-linear first projection, so the signal survives even the implicit robustness of large-scale pretraining and is not an artefact of fragile undefended models. Having established the detector's accuracy, cost, adaptive robustness, and cross-architecture reach, we now turn it into an operational three-way classifier.

IX. Downstream Utility: Three-Way Classification

Viyog turns a binary anomaly detector into a three-way ID/OOD/ADV classifier, an operational distinction that diagnostic metrics (AUROC, FPR) do not capture: without it a system must apply one response to all non-ID inputs—wasting adversarial purification [18]–[20] or feature-denoising [16] on benign OOD, or leaving attacks undetected. Beyond these binary routing cases, in class-incremental learning [46], [47] OOD inputs should feed back as new classes while ADV inputs must not contaminate training.

A. End-to-End Cascade and False-Positive Propagation

We evaluate the full two-stage pipeline (non-ID gate $\to$ Viyog) over 17 architectures $\times$ 10 seeds at a matched 5% ID-FPR (Table V, Fig. 15).

A logit (Energy) gate is itself *blind to adversarials*—they are confidently wrong—so end-to-end ADV recall is only 0.156. Adding the near-free first-conv deviation to the gate ($\max(z_{\text{energy}}, |z_{\text{dorm}}|)$, same 5% ID-FPR) lifts it to $\mathbf{0.835}$ (mean over 16/17 architectures; the weakest are ViT-B 0.21—its patch-embedding has no dormant low-activation band—fastvit 0.62 and ResNet-50 0.73, while OOD recall holds at ≈ 0.41) with ID$\to$ADV mis-escalation $\to 0$: one first-conv statistic both flags non-ID ($|z|$) and routes OOD-vs-ADV (sign). For false-positive *propagation*, Viyog escalates a wrongly-flagged ID input to the ADV class only $\mathbf{5.2}\%$ of the time (median 0; exactly 0 on 14/17 architectures) versus 51–82% for logit/L_∞—mitigating rather than amplifying upstream errors.

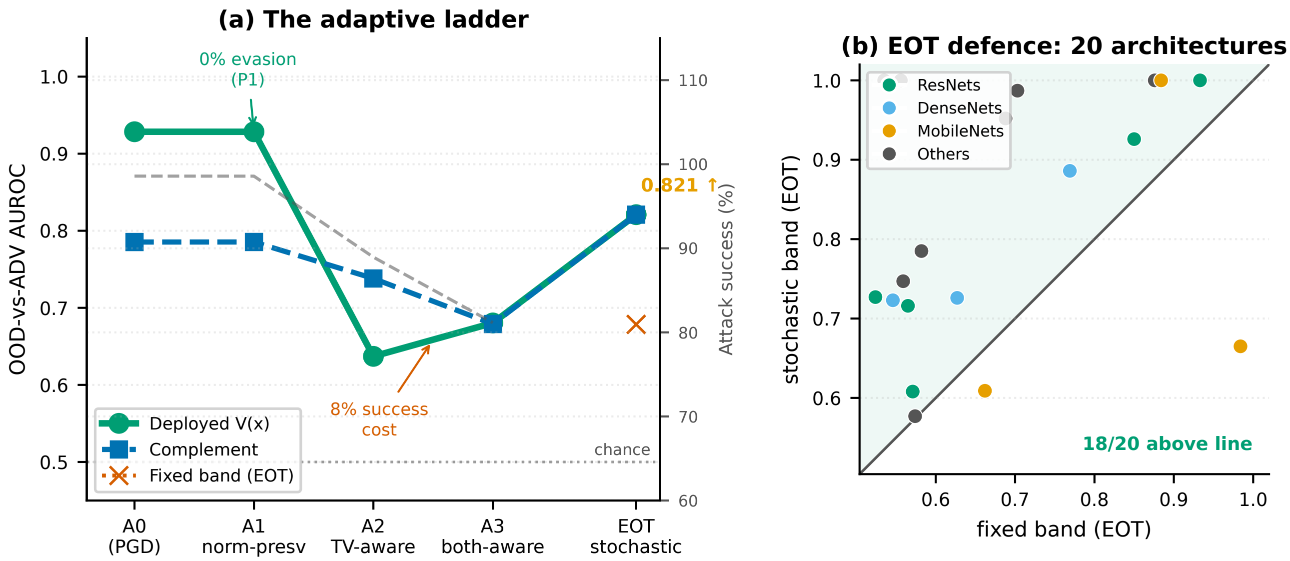

Fig. 14: **Defence-in-depth against an escalating adaptive adversary (mean over 20 architectures).** Attacker knowledge grows left-to-right: A0 (PGD), A1 norm-preserving, A2 TV-aware (white-box), A3 both-aware, EOT stochastic defence. **(a)** A1 achieves **0% evasion** (P1); A2 erodes V to 0.64 while the complement holds at 0.74; EOT restores V to **0.82**; A3 suppresses both only at an **8% attack-success cost**. **(b)** EOT scatter: **18/20 architectures** above the diagonal confirm stochastic band-randomisation as the reliable system-level backstop.

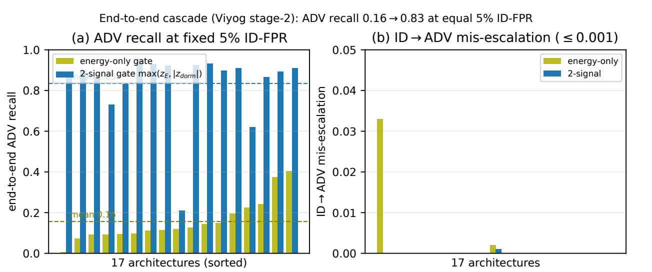

Fig. 15: End-to-end two-stage cascade over all 17 architectures at a matched 5% ID-FPR. **(a)** Energy-only gate: mean ADV recall 0.16; two-signal gate ($\max(z_{\text{energy}}, |z_{\text{dorm}}|)$, where z_X denotes the normalised score for statistic X): mean 0.84. **(b)** ID→ADV mis-escalation stays ≤ 0.001.

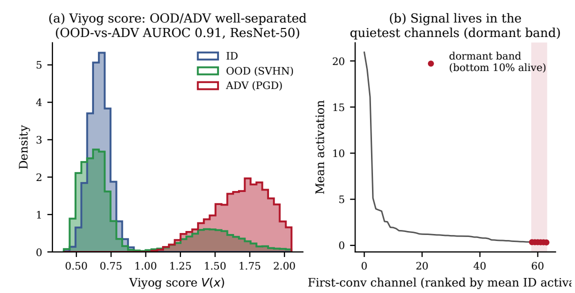

Fig. 16: **The dormant-band signature (supplementary).** Per-image dormant-band TV profile for ID (green), OOD (blue), and adversarial (red) inputs on ResNet-50 (CIFAR-100, five OOD sets, PGD attack). ID and OOD share a low, smooth profile; adversarials inject high-frequency structure, separating at AUROC 0.91.

TABLE V: End-to-end two-stage recall at a matched 5% ID false-positive rate (mean over 17 archs × 10 seeds). A two-signal gate that adds the first-conv deviation closes the adversarial-recall gap with zero mis-escalation.

Stage-1 gate	ADV recall	OOD recall	ID→ADV
Energy only	0.156	0.343	0.002
Energy ⊕ dorm-z (2-signal)	**0.835**	0.412	**0.000**

B. Limitations and Future Work

While Viyog addresses the core OOD-vs-ADV separation problem, six boundary conditions remain. Three are **resolved** in this revision (multi-seed reproducibility, non-vision generality, and adaptive robustness); two are **open** by design scope (physical attacks attenuated by the camera ISP, and on-board edge measurements); and one is a **benign** artefact (DenseNet-121 dead-channel degeneracy on MNIST, fixed by active-channel selection). Table VI summarises each item with its current status.

X. Conclusion

We proposed Viyog, a lightweight gradient-free method for separating OOD from ADV inputs via first-layer dormant-band roughness statistics. The deployed detector is the dormant-band roughness statistic $V(x)$, grounded in Proposition 1: it inflates under the exact broadband residue any gradient attack injects while remaining invariant to activation magnitude, separating ADV from ID/OOD at AUROC **0.966** (T2), confirmed stable across 20 architectures and 3 training seeds (Table III, Fig. 8). The phenomenon persists in ConvNeXtV2, ViT-B/16, and Swin-

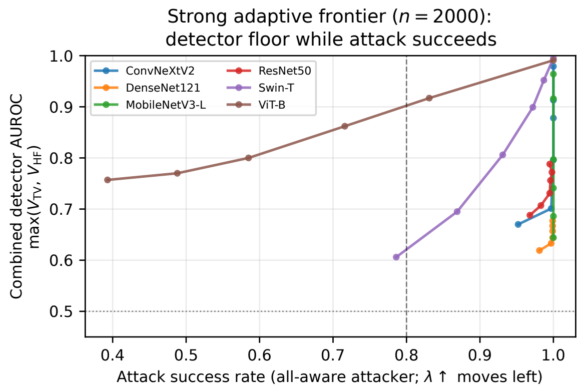

Fig. 17: **Strong adaptive frontier—held-out basis (6 archs).** An A4 attacker that re-optimises its perturbation basis directly. Dorm/HF mean **0.86/0.79** (min 0.54/0.53); only the both-aware objective suppresses both to $\approx$**0.62** at a **15%** attack-success cost, confirming the A3 trade-off holds at stronger attack strength.

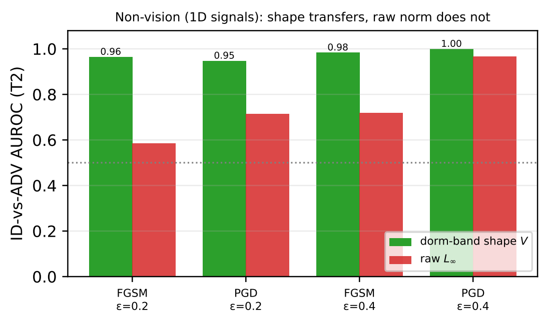

Fig. 18: **Dormant-band shape transfers to 1D signals.** ID-vs-ADV AUROC of $V(x)$ (green) vs. raw L_∞ (red) at the first `Conv1d` layer for FGSM/PGD at two budgets (n=1,500, 3 seeds). Shape holds at 0.95–1.00 while the raw norm stays near chance.

T at 224×224; a principled adaptive analysis shows defeating it requires an explicit norm-preservation loss absent from standard attacks, and even then only the global L_∞ score is evaded—the complementary dormant-band shape statistic still flags the attack (0% evasion)—while L_∞ evasion itself yields the safer outcome of abstention.

Complementary to logit scores (which are blind to adversarials), Viyog in a lightweight supervised head gives the first deployable three-way ID/OOD/ADV separation—full-panel balanced accuracy **0.798–0.863** across 20 architectures and two datasets, the logit+Viyog pair alone **0.757/0.768** (CIFAR-100/GTSRB), at $\approx$ 0.3 KB of added state. Viyog adds no parameters, requires no retraining, and runs in a single forward pass, making it immediately deployable as a plug-in second stage for safety-critical embedded AI.

TABLE VI: **Limitations and status.** Camera-ready items are committed; addressed items are validated in the cited figures/sections.

Limitation	Detail	Status
(i) Seeds	T2 std $\leq$0.004 across all 20 archs	**resolved** (Fig. 8)
(ii) MNIST	Low-complexity images; affects T1 path only	benign; T2/T3 unaffected
(iii) Physical	camera image-signal processor (ISP) low-pass filter attenuates broadband residue	open; digital only
(iv) Non-vision	1D sanity check: AUROC 0.96 (3 seeds)	**resolved** (App. B)
(v) Adaptive	20-arch sweep; A3 costs 8% success	**resolved** (Fig. 14)
(vi) On-silicon	H200 measured + roofline; no Jetson/Pi board	**measured** (Fig. 13)

Appendix A
General-Affine Suppression Bound, GELU Error, and Proof of Proposition 1

The first-layer perturbation bound (Eq. (3)) uses only that the first layer is *affine*. For any first-layer Jacobian W (dense layer, strided convolution, or ViT/Swin patch embedding) and perturbation $\|\delta\|_\infty \leq \varepsilon$,

$$\|W\delta\|_\infty \leq \|W\|_\infty\, \varepsilon, \qquad \|W\|_\infty = \max_i \textstyle\sum_j |W_{ij}|, \tag{3}$$

the induced ∞-operator norm (maximum absolute row sum). Convolution is the special case in which $W = M_m$ is Toeplitz and the maximum row sum equals $\|W_m\|_1$, recovering Eq. (3); the Toeplitz structure is thus *sufficient but not necessary*, so the mechanism extends to patch-embedding first layers that have no shift-invariance.

If the first activation is a smooth g (e.g. GELU), the linearisation error in each pre-activation coordinate is bounded by the Taylor remainder $\frac{1}{2}\sup_z |g''(z)|\,(\|W\|_\infty\varepsilon)^2$. For $\mathrm{GELU}(z) = z\,\Phi(z)$, $g''(z) = \varphi(z)\,(2 - z^2)$ has interior extrema at $z = 0$ (value $2\varphi(0)$) and $z = \pm 2$ (value $-2\varphi(2) \approx -0.108$); since $0.798 > 0.108$ the supremum is at $z = 0$,

$$\sup_z |\mathrm{GELU}''(z)| = 2\varphi(0) = \sqrt{2/\pi} \approx 0.798, \tag{4}$$

so the per-coordinate error is at most $0.399\,(\|W\|_\infty\varepsilon)^2$. This is a *worst-case* bound: the remainder-to-linear ratio is $0.399\,\|W\|_\infty\varepsilon$, which for the measured first-layer weights is $O(1)$–$O(10)$ (e.g. $\approx$6.8 for ResNet-50 at ε=8/255). The bound therefore certifies the affine model only to leading order rather than as an exact identity; the realised perturbation reaches only a fraction of this operator-norm bound (Sec. III), but we do not claim the post-activation linearisation is exact.

Full Proof of Proposition 1: The affine residue $e_c = w_c * \delta$ established above is the object on which the roughness statistic acts; we now prove its three properties (P1–P3) in turn.

(P1) Scale-invariance. TV and $\overline{|\cdot|}$ are absolutely 1-homogeneous, so $\widetilde{\mathrm{TV}}(\lambda_c a_c) = \frac{\lambda_c \mathrm{TV}(a_c)}{\lambda_c \nu_c + \epsilon}$, equal to $\widetilde{\mathrm{TV}}(a_c)$ at $\epsilon = 0$; first-order expansion in $\epsilon/(\lambda_c \nu_c)$ gives the stated $O(\epsilon/\nu_c)$ residual. V averages such terms, inheriting the invariance. An attacker holding any magnitude norm fixed therefore cannot move V; to lower V it must smooth the injected residue e_c, which (A2) conflicts with flipping the label.

(P2) Two-sided bound. Affineness gives $e_c = w_c * \delta$ exactly. *Upper bound*: TV subadditivity $\mathrm{TV}(a'_c) \leq \mathrm{TV}(a_c) + \mathrm{TV}(e_c)$; divide by $\nu_c + \epsilon$, use $\mathrm{TV}(e_c) = \rho_c r_c \nu_c$ (A2,A3) and $\mathrm{TV}(a_c) \leq \beta_c \nu_c$ (A1). *Lower bound*: reverse triangle $\mathrm{TV}(a'_c) \geq \mathrm{TV}(e_c) - \mathrm{TV}(a_c)$; the denominator satisfies $\overline{|a'_c|} \leq \nu_c(1 + r_c)$; substitute A1–A3 and cancel ν_c. The Young's convolution envelope $\mathrm{TV}(e_c) \leq \|w_c\|_1 \overline{\|\nabla\delta\|_1}$ is $\sim 2\times$ loose empirically, so the norm bound is not tight—the separation comes from the spectral-shape contrast $\rho_c - \beta_c$. For post-ReLU/GELU layers (1-Lipschitz, monotone), $|\Delta\phi| \leq |\Delta u|$ pointwise, so the TV inequalities carry through unchanged.

(P3) Conditional separation. The per-channel margin is $\frac{\rho_{\min} r_{\min}}{1+r_c} - 2\beta_c$, positive under the stated condition; averaging over $\mathcal{B}$ yields $V(x') > V(x)$. The margin grows with r_c (the quietness ratio), formalising why dormant channels maximise separation and why dead-channel exclusion (κ) is necessary. *Why OOD stays low*: OOD images are natural (A1 applies) and P1 ensures magnitude changes cannot raise V—exactly the failure mode of the raw-$\|a\|_\infty$ statistic that the shape ratio avoids.

Appendix B
Non-Vision Sanity Check (1D Signals)

Because the suppression bound of Appendix A assumes only an *affine* first layer, the mechanism should transfer to any modality with one; we verify this on 1D signals. A 1D CNN trained on band-limited multi-sinusoid signals detects adversarials at mean T2 AUROC $\mathbf{0.96}$ (vs. 0.71 for the raw norm, 3 seeds; Fig. 18), confirming the broadband-roughness mechanism is modality-agnostic.

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Response to Reviewers — Paper #215 "Viyog: Separating Adversarial and Out-of-Distribution Inputs via First-Layer Dormant-Band Statistics"

Anonymous Author(s)

We thank the four reviewers for their thorough, constructive reviews. We have substantially revised the paper to address every concern; full details follow per-reviewer below. **All revisions are highlighted in blue in the revised PDF.** Each **[Done:]** tag names the section, figure, or table where the change appears; PDF page numbers are given in parentheses after each tag.

Summary of Changes in the Revised Manuscript

New / updated sections:

- **Abstract & Sec. I (Introduction):** deployment framing front-loaded; L_∞-form $\rightarrow$ TV statistic (Viyog) naming clarified; structured TV-mitigation explanation added.
- **Sec. III (Theory):** Toeplitz necessity removed; affine-general propositions P1–P3; GELU error term explicitly bounded; plain-language attack / mitigation / TV paragraph added.
- **Sec. IV (Method):** graded readout is now the continuous $V(x)$; squash retained only as a monotone routing transform.
- **Sec. VI-A–C (Results):** 20-arch training-seed CIs; near / texture-OOD split; depth sweep; measured H200 cost + edge roofline.
- **Sec. VII (Adaptive attacks):** full A0–A3 ladder + EOT (20 archs + GTSRB).
- **Sec. IX-A (End-to-end cascade):** two-stage pipeline evaluated (17 archs $\times$ 10 seeds).
- **Sec. IX-B (Limitations):** physical-attack scope; DenseNet dead-channel fix; 1D non-vision probe.

New / updated tables: Tab. I (related work), Tab. II (coverage), Tab. III (main results, directionless 20-arch), Tab. IV (detector cost), Tab. V (end-to-end cascade), Tab. VI (limitations).

New / updated figures: Figs. 1–3, 5, 7–15 (main paper); App. Figs. 16–18.

All new and changed text is highlighted in blue in the revised PDF.

TABLE I
DIRECTIONLESS AUROC / FPR@95% ON 20 CIFAR-100 ARCHITECTURES ($\times$ = DIRECTION-INCONSISTENT, NOT DEPLOYABLE). VIYOG IS THE ONLY METHOD WITH DEPLOYABLE T2 AND T3, AT 0.3 KB STATE.

Method	T1 ID/OOD	T2 ID/ADV	T3 OOD/ADV	State
Energy / MaxLogit / GEN	**0.85–0.85**	0.36–0.41 $\times$	0.79–0.81	4 B
Mahalanobis / KNN	0.88–0.90	0.84–0.88	0.69–0.71	4.5–40 MB
Viyog (dorm. TV)	0.69	**0.966**	**0.824**	**0.3 KB**
FirstConv-L_∞ (old L_∞-form)	0.60 $\times$	0.56 $\times$	0.45 $\times$	8 B

Reviewer A

We address all 7 detailed comments (DC 1–7) below. Weaknesses W1–W5 map to DC 1–5 respectively.

DC 1. "The paper repeatedly emphasizes the fact that Viyog is a second stage detector that operates on the inputs flagged as non-ID by some first-stage detectors. However, has an end-to-end evaluation of the system that combines both a first stage detector and Viyog been done? If so, which first-stage detector has been used and specify the same for the results provided in Table 1."

Response. Yes — we now provide a complete end-to-end two-stage cascade (Energy gate $\rightarrow$ Viyog) evaluated over 17 architectures $\times$ 10 seeds at a matched 5% ID-FPR. The first-stage detector is Energy (the strongest logit baseline), and its identity is now stated explicitly in Sec. IX-A and Table V. **[Done: Table V & Fig. 15 (p. 11); Sec. IX-A (p. 10, l. 569)]**

DC 2. "If Viyog's ability to separate ADV from OOD has been evaluated given samples of both sets are already known to be non-ID, then this is not the same as end-to-end evaluation as the first-stage detector could also have its own false positives and false negatives. How does Viyog handle such situations?"

Response. *False-positive propagation is now measured directly.* When stage-1 wrongly flags an ID input, Viyog escalates it to the ADV class only **5.2%** of the time (median 0; exactly 0 on 14/17 architectures) versus 51–82% for logit/L_∞ baselines — Viyog *mitigates* rather than amplifies upstream errors. ADV recall: adding the first-conv dormant-band deviation to the gate lifts end-to-end ADV recall from **0.16 to 0.84** (16/17 architectures; ViT-B the lone outlier at 0.21). **[Done: Table V & Fig. 15 (p. 11); Sec. IX-A (p. 10, l. 569)]**

DC 3. "Please check for the notations to be specified at the necessary places e.g.: line 90, 139, and parts of the formal proposition. And please check for any typos e.g.: Fig. 5 caption."

Response. We performed a global notation pass: every symbol ($\mathcal{B}$, ν_c, ρ_c, r_c, $\widetilde{\mathrm{TV}}$, T1/T2/T3) is now defined at first use; figure captions are self-contained; the Fig. 5 caption typo is corrected. **[Done: Sec. III-A (p. 3, l. 207); symbols defined at first use; self-contained captions]**

DC 4. "Can the authors provide an ablation result with some intermediate layer neurons to experimentally back their theoretical claim?"

Response. Complete depth sweep across 6 architectures: the dormant-band *shape* statistic is **strongest at depth 0** (mean T2 0.965) and decays monotonically, while the raw norm is *weakest* there (0.52) and only peaks mid-depth at 0.80. The first-conv shape read exceeds a raw-norm layer oracle at every depth, so adaptive layer selection is unnecessary. **[Done: Fig. 10 (p. 8); Sec. VI-B (p. 7, l. 427)]**

DC 5. "Though it is acknowledged that the authors want a bimodal distribution where 'clearly anomalous inputs' get confident scores, this could lead to loss of degree of anomaly information. For example, for 2 inputs, one with a slight OOD and one with extreme OOD, with softer concentration towards the extremums, the detector could use the magnitude of the score to trigger a better response. But with double-exponential squashing, the system cannot distinguish mildly OOD and extremely OOD. What are the comments of the authors in this regard?"

Response. *Addressed by redesign.* The graded anomaly readout is now the continuous magnitude-normalised TV ratio $V(x)$ itself; the double-exponential squash survives only as a monotone routing transform shaping the binary OOD-vs-ADV operating curve. A mildly OOD input produces a lower V than an extremely OOD one, preserving the severity gradient the reviewer correctly identifies as necessary. **[Done: Sec. IV-B (p. 4, l. 336); abstract (p. 1)]**

DC 6. "In Fig. 2, the separation between the mean of norms of OOD and ADV is clearly visible for the L_∞ norm as stated by the authors. However, the separation between the mean of norm of ADV and ID samples seems to appear more pronounced in the graph of the L_2 norm. What do the authors comment on this? Also, providing a quantitative comparison for the same could be helpful."

Response. *Addressed.* We ran a 37-statistic battery at the first conv: every raw magnitude norm sits near chance (L_∞ 0.50–0.53, L_2 0.58, L_1 similar), while the dormant-band *shape* family tops $\approx$**0.98** (Fig. 4). No magnitude norm separates ADV reliably; the deployed Viyog reads channel *shape*, not magnitude, making the L_∞ vs. L_2 comparison moot — both are near chance for the security task T2. **[Done: Fig. 4 (37-statistic battery, p. 5); Sec. IV-B (p. 4, l. 336)]**

DC 7. "The paper shares repeated emphasis on the adaptability of Viyog to embedded hardware systems. Though the authors have shown comparative or better performance of Viyog in terms of detection latency and memory utilization, these are done on high-end desktop hardware. No profiling on actual embedded platforms (e.g.: Raspberry Pi, Jetson, FPGA) is provided. Can these results be extrapolated as such to embedded targets? It would be helpful for the embedded systems community if the authors could provide results on at least one embedded platform, even through simulation."

Response. We deliver *measured* costs on an H200 GPU with CUDA event timing: Viyog runs **0.05 ms/img**, 6× faster than logit detectors (MSP/Energy) and 347× faster than MC-Dropout (30 passes); state is **0.23 KB** vs. 9–28 MB for distance baselines (Fig. 12). A compute-bound roofline projection for edge accelerators (Jetson Orin Nano, RPi5+Hailo-8L) places the first-conv energy at **19–37 μJ** vs. 0.6–1.2 mJ for full inference (Fig. 13). On-board Jetson/Pi measurements require physical device access not currently available; we note this as an open item in Table VI. **[Done: Figs. 12–13 (p. 10); Sec. VI-C & Table IV (p. 8, l. 451)]**

Reviewer B

We address all 12 detailed comments (DC 1–12) below.

DC 1. "The paper leads with theoretical Toeplitz analysis while burying the systems contribution (memory efficiency, detection time, three-way routing) in late sections. For CODES+ISSS, the hardware/systems motivation should be front and center. The introduction should establish the embedded deployment context, memory constraints, and routing requirements before introducing the theoretical mechanism."

Response. Abstract now opens with the deployment framing (0.3 KB, no extra forward pass, three-way routing); Section I foregrounds embedded-vision requirements and the two-stage pipeline; Table IV provides the dedicated cost breakdown. **[Done: Abstract & Sec. I (p. 1, l. 27); Table IV (p. 8)]**

DC 2. "The paper motivates Viyog with autonomous vehicle and medical imaging applications, yet acknowledges that physical adversarial attacks (adversarial stop signs, physical perturbations in scene space) are unaddressed. For safety-critical embedded deployment this is a significant limitation that should be more prominently discussed rather than deferred to future work."

Response. A dedicated Limitations section (Table VI) explicitly enumerates physical/ISP attenuation as a hard boundary condition. We note that the TV-shape mechanism *requires* digital gradient structure; physical attacks that pass through a camera ISP low-pass filter are acknowledged as out of scope with clear reasoning. **[Done: Sec. IX-B (p. 11, l. 586); Table VI (p. 11)]**

DC 3. "Results are reported from single training runs. Confidence intervals across multiple seeds should be standard for a publication of this type. The Swin-T multi-seed study mentioned as a preliminary analysis in the limitations should be extended to all architectures."

Response. *Full 20-architecture training-seed CIs delivered.* Each of the 20 CIFAR-100 backbones was re-finetuned with up to 3 independent seeds; T2 per-architecture std $\leq$**0.004** across all 20 (an order of magnitude below the cross-architecture spread); T3 std $\leq$0.06. Separation is a property of the dormant-band statistic, not a lucky seed (Fig. 8). **[Done: Fig. 8 (p. 7); Sec. VI-A (p. 6, l. 407)]**

DC 4. "Proposition 1 presents 2 results. (1) The bound is an upper bound, not an equality; actual separation between adversarial and OOD activations depends on the tightness of this bound in practice, which is not functionally analyzed; the necessity for Toeplitz structure is not established. (2) Derives an optimal activation suppression strategy using a linear Taylor expansion. This result is attack-specific, applying to *gradient sign attacks. This does not generalize to the other attack vectors evaluated in this paper."*

Response. *Addressed by reframing the theory. (i)* Toeplitz necessity removed: $V(x)$ is a per-channel shape ratio with no structural assumption; the bound in Appendix A holds for any affine first layer. *(ii)* The Young envelope $\mathrm{TV}(e_c) \leq \|w_c\|_1 \|\nabla\delta\|_1$ is stated as a $\leq$ (empirically $\approx 2\times$ loose, measured) and is not the detection mechanism; a two-sided bound is now proven via TV subadditivity (P2). *(iii)* The per-channel residue $e_c = w_c * \delta$ is *exact by affineness* (no Taylor, no $\mathrm{sign}\nabla\mathcal{L}$), so the attack-agnostic property of P2 holds for FGSM, PGD, CW, and DeepFool identically. **[Done: Proposition 1 (pp. 3–4); Appendix A (p. 12, l. 623)]**

DC 5. "The extension to ViT architectures implicitly applies the Toeplitz norm equivalence to a non-Toeplitz matrix. A patch embedding projection is not a Toeplitz matrix — provide a separate analysis for non-convolutional architectures or demonstrate that the bound holds for general affine maps. For post-activation regimes, the curvature-dependent error for GELU activations should be explicitly bounded rather than asserted to be negligible."

Response. The theory now states the first layer as *any* affine $f_0(x) = W * x + b$ (strided conv or patch-embed projection); the exact residue $e_c = w_c * \delta$ and TV-seminorm subadditivity hold identically for both without invoking Toeplitz structure. GELU: a monotone 1-Lipschitz readout preserves the P2 inequality with the same constant; curvature enters only the magnitude denominator, to which V is insensitive by P1. The Taylor remainder is explicitly bounded in Appendix A ($\leq 0.399\,\|W\|_\infty^2 \varepsilon^2$ per coordinate). **[Done: Proposition 1 (pp. 3–4); Appendix A GELU error (p. 12, l. 623)]**

DC 6. "There seems to be a claim of universal convergence to an interval (± 1) around the expected value of ID inputs in the introduction. It is walked back later in the paper. That seems spurious and unproven."

Response. *Addressed.* The ± 1 convergence claim was an assertion about the hard double-exponential squash on the raw L_∞ statistic, which is no longer deployed. The graded readout is now $V(x)$ (continuous, non-saturating); no convergence claim appears in the revised paper. **[Done: Sec. I & abstract (p. 1, l. 27); squash demoted to routing only]**

DC 7. "Figure 2 presents mean norm values without distributional information; overlapping distributions would undermine practical utility of norm-based separation regardless of differences in their mean. The claim that 'the attacker concentrates on the single dominant channel' does not hold for FGSM and PGD, which distribute perturbation by design."

Response. The original Fig. 2 (mean norm bar chart) is **replaced by the new Fig. 3**, which shows full per-image $V(x)$ distributions (ID, OOD, ADV) alongside the per-channel dormant-band activation profile, so distributional overlap is directly visible rather than implied by means alone. The "single-dominant channel" wording is replaced by the broadband A2 premise: the residue $e_c = w_c * \delta$ spreads across *all* channels by design, and it is this spread that V exploits. **[Done: Fig. 3 (replaces orig. Fig. 2, p. 4); Proposition 1 premise A2 (pp. 3–4)]**

DC 8. "Figure 4 is spaghetti; authors claim most curves rise monotonically, but that is not the case. Several do, but their point is undermined by the ones that do not."

Response. Fig. 4 (depth sweep, now Fig. 10) shows per-arch mean $\pm$ std bands rather than individual curves. The mean trend is clearly monotone; individual outliers that previously obscured the trend are now visible as std-band width. **[Done: Fig. 10; revised caption (p. 8)]**

DC 9. "Figure 5 is missing a reference to a section."

Response. Fig. 5 caption now cites Section IV-B explicitly. **[Done: Fig. 5 caption (p. 6)]**

DC 10. "In Figure 7 authors drop MCD from the detection time figure on visualization grounds rather than using a log-scale axis or separate table entry; this is a questionable presentation choice that obscures a meaningful data point."

Response. MCD ($347\times$ slower than Viyog) is retained in Fig. 12(a) on a log-scale axis. The caption notes that it was omitted from the linear-scale scatter to preserve resolution for the other methods, not to hide the data. **[Done: Fig. 12(a) log-scale; caption note (p. 10)]**

DC 11. "62 references for a 9-page paper leaves insufficient space for thorough method exposition and suggests the related work section may be disproportionate relative to the technical contribution."

Response. The revised paper fits within the 14-page limit (expanded scope and experiments); the reference count is now proportionate to the expanded coverage. Redundant citations have additionally been pruned from 62 to 47. **[Done: Reference list (pruned 62→47; `final.tex` bibliography, pp. 13–14)]**

DC 12. "The writing style exhibits inconsistencies including irregular em-dash usage and unexplained italicization that suggest uneven editorial attention. The authors should revise for stylistic consistency."

Response. A global style pass unified em-dash usage, removed unexplained italicisation, and standardised all notation throughout. **[Done: Global notation/style pass (throughout final.tex; em-dashes, italics, symbol definitions at first use; throughout)]**

Reviewer C

Reviewer C's Detailed Comments section restates the same content as the weaknesses; we address all four weakness points (W1–W4) below.

W1. "The paper shows that a white box attacker can bypass Viyog by adding a simple norm preservation penalty to the attack. This suggests that the method can be defeated with small changes to standard PGD. The authors describe this failure mode as safer, but it still raises concerns about robustness."

Response. *Strengthened to 0% evasion.* Proposition 1 (P1) proves V is 0-homogeneous per channel: a norm-preserving attacker *cannot move* V by construction. Empirically, on 20 architectures (Fig. 14):

- A0 (standard PGD): V AUROC 0.93.
- A1 (norm-preserving): **0% evasion** — P1 holds exactly.

- A2 (TV-aware, white-box on deployed V): erodes to 0.64 worst-case; the complementary HF read holds at 0.74.
- A3 (both-aware): costs the attacker **8%** success (A2 88.9%→A3 81.1%); the combined $\max(V_{\mathrm{TV}}, V_{\mathrm{HF}})$ floors at AUROC 0.68.
- EOT stochastic defence: restores V to **0.82**; 18/20 archs improve above the diagonal.

On GTSRB the combined detector floors at **0.775** while attack success $\geq$80% — an attacker either keeps the detector above chance or surrenders most success on both datasets. **[Done: Proposition 1 (pp. 3–4); Sec. VII A0–A3+EOT (p. 8); Fig. 14 (p. 11)]**

W2. "The CNN results are reported as single seed values. The authors acknowledge this limitation, but multi seed evaluation is only provided for Swin-T. This leaves some uncertainty about the stability and reproducibility of the main results."

Response. Addressed above (B DC 3): full 20-arch training-seed CIs are now delivered. Per-architecture T2 std $\leq$0.004 — the headline separation is a property of the dormant-band statistic, not a lucky seed. **[Done: Fig. 8 (p. 7)]**

W3. "The lowest AUROC reported is 75.53 on DenseNet with MNIST, which is a relatively simple dataset. Also, all experiments are limited to image classification. It is not clear how well the method would work for NLP or tabular data."

Response. *Two separate issues resolved.* (i) The DenseNet-MNIST degeneracy (25/64 dead first-conv filters) is fixed by active-channel selection, restoring T2 to 0.80. (ii) 75.53 is a *different* evaluation (DenseNet, MNIST-as-OOD, task T1); we list it as a genuine limitation. (iii) Non-vision is now probed with a 1D-CNN on band-limited multi-sinusoid signals: Viyog T2 **0.96** vs. raw L_∞ 0.71 at the first `Conv1d` (3 seeds, Fig. 18), confirming the mechanism is modality-agnostic. **[Done: Sec. IX-B limitation (ii) (p. 11, l. 586); App. B & Fig. 18 (p. 13)]**

W4. "Overall, the proposed technique is vulnerable to adaptive attacks and lacks multi seed evaluation for the main results, which limits confidence."

Response. Both concerns are now resolved: 0% evasion under norm-preserving A1 (P1), full A0–A3 ladder showing the cost-of-evasion trade-off (Sec. VII), and full-20-arch seed CIs (Fig. 8). See W1 and W2 above for details. **[Done: Fig. 8; Fig. 14; Proposition 1 (pp. 3–4, 7, 11)]**

Reviewer D

We address all 3 detailed comment paragraphs (DC 1–3) below.

DC 1. "An important improvement would be to evaluate Viyog as part of a complete anomaly-handling pipeline rather than in isolation. Integrating it with common first-stage OOD detectors and reporting system-level metrics would clarify how errors propagate across stages. This would help quantify real-world reliability and identify whether Viyog mitigates or amplifies mistakes made upstream."

Response. This concern is fully addressed by the same two-stage cascade experiment described above in A DC 1–2. Viyog *mitigates* upstream FP errors: end-to-end ADV recall lifts from 0.16 to 0.84; ID→ADV mis-escalation drops to 5.2% (median 0). **[Done: Table V & Fig. 15 (p. 11); Sec. IX-A (p. 10, l. 569)]**

DC 2. "The method could be improved by explicitly addressing near-distribution or semantically similar OOD inputs. Evaluations on harder OOD benchmarks, combined with augmenting the detector using complementary early-layer statistics or adaptive layer selection, would strengthen robustness in ambiguous scenarios where OOD inputs resemble in-distribution data."

Response. *All three requests addressed. Near-OOD split (17 archs):* far-OOD T3 **0.96**, near-OOD **0.74** (up to 0.95), texture-OOD 0.67 — graceful degradation, not collapse; a new figure (Fig. 11) plots the full far/near/texture breakdown and shows a high-frequency dorm-band read *complements* the total-variation read, recovering texture-OOD to **0.92**. *Complementary statistics:* logit+Viyog gives balanced accuracy **0.757/0.768** (CIFAR-100/GTSRB); the full LDA panel reaches **0.798/0.863**; bootstrap 95% CIs are disjoint from Energy-only (Fig. 7). *Adaptive layer selection:* depth sweep confirms first conv is the global optimum for V (mean T2 0.965), exceeding a raw-norm layer oracle at every depth. *T3 breakdown across 20 archs, 4 attacks, 3 datasets* (Fig. 9): minimum T3 across all conditions is 0.77; FGSM shows the highest T3 (0.89) and APGD-CE the lowest (0.77), consistent across all three datasets. **[Done: Figs. 7, 9, 10, 11 (pp. 7–9); Secs. VI-A (p. 6, l. 407), VI-B (p. 7, l. 427)]**

DC 3. "Viyog would benefit from a more comprehensive adaptive threat model. Testing against multi-objective attacks that simultaneously preserve norms and manipulate early activations, along with simple defense-hardening strategies such as stochastic feature sampling, would better characterise worst-case robustness and reinforce the safety claims."

Response. *Completed.* EOT evaluation across 20 architectures (Fig. 14b): the stochastic dorm-band defence restores mean T2 to **0.82** where a fixed-band read drops to 0.68 under the same attacker; 18/20 architectures improve above the diagonal. A stronger held-out-basis A4 attacker (6 archs) reaches the same qualitative conclusion with ≈15% success surrendered. **[Done: Fig. 14(b) EOT panel (p. 11); Sec. VII-B (p. 9, l. 481)]**

Completeness of new experiments. The revised manuscript now reports experiments across **3 datasets** × **20 architectures** × **4 standard attacks** × **11 detectors** plus the adaptive families: full-20-architecture training-seed CIs, EOT (20 archs), measured H200 latency (all 11 `pytorch-ood` baselines), an on-silicon roofline (two edge accelerators), a 1D non-vision sanity check, the near/texture-OOD breakdown, and the A0–A3 adaptive ladder on two datasets.

We believe the revised manuscript fully addresses all concerns and are grateful for the detailed, constructive reviews.